Agenda

1. Sets ✓
2. Dictionaries
3. conditional statement
4. loops — for
   — while.

$(1, 1, 3, 4)$

Sets

collection of unique items

$x = \{1, 2, 3, 4\}$

A set is like a bag where every item appears only once

$x = \{2, 1, 3, 6\}$

Unordered

List = [2, 2, 3, 4, 5]

- change ✓
- Repeated ✓
- ordered ✓

tuples = (2, 2, 3, 4, 5) → Change ✗

- Repeated ✓
- ordered ✓

Sets = {2, 3, 5, 4}

- change ✓
- Repeated ✗
- Ordered ✗

Where are the Sets used ?

→ Remove duplicate data

akshat
John
akshat
Iman

data
login

10,000

names = ["akshat", "John", "akshat"]

unique_names = set(names) ← convert the names (list) to the sets.

print(unique_names) ⟹ {"akshat" "John"}

set

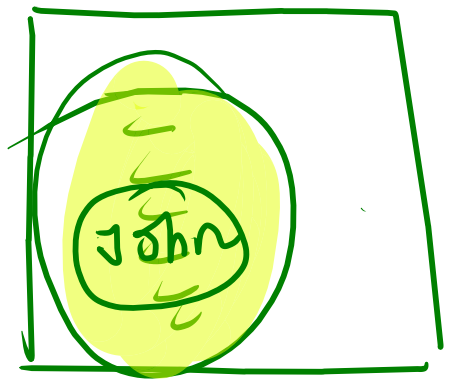
unique values

② Databases & unique Records

↳ Ensure no duplicate values

user emails must be unique

Wallmart → Premium kids product



set 1

set of data 1

set 2

set of data 2

People who purchased Diapers on 6th march

People whose cart value > $1000

I want to find the people who are in set 1 & set 2

Intersection in sets

Basic Operations in sets

① **Union**

$A = \{1, 2, 3\}$

$B = \{3, 4, 5\}$

$A \cup B = \{1, 2, 3, 4, 5\}$

② **Intersection**

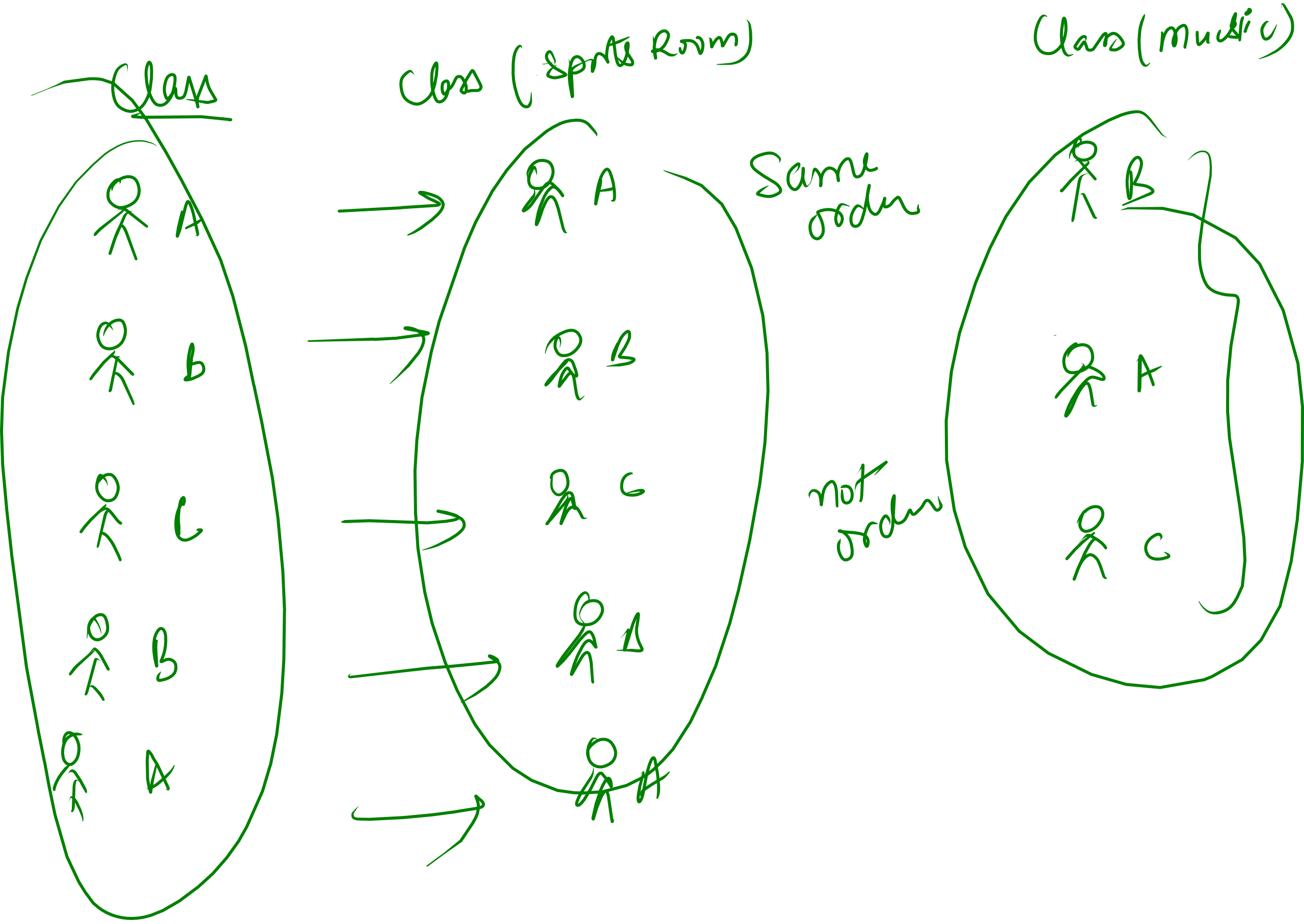
$A \cap B = \{3\}$

③ Difference ⇒ $A - B = \{1, 2\}$

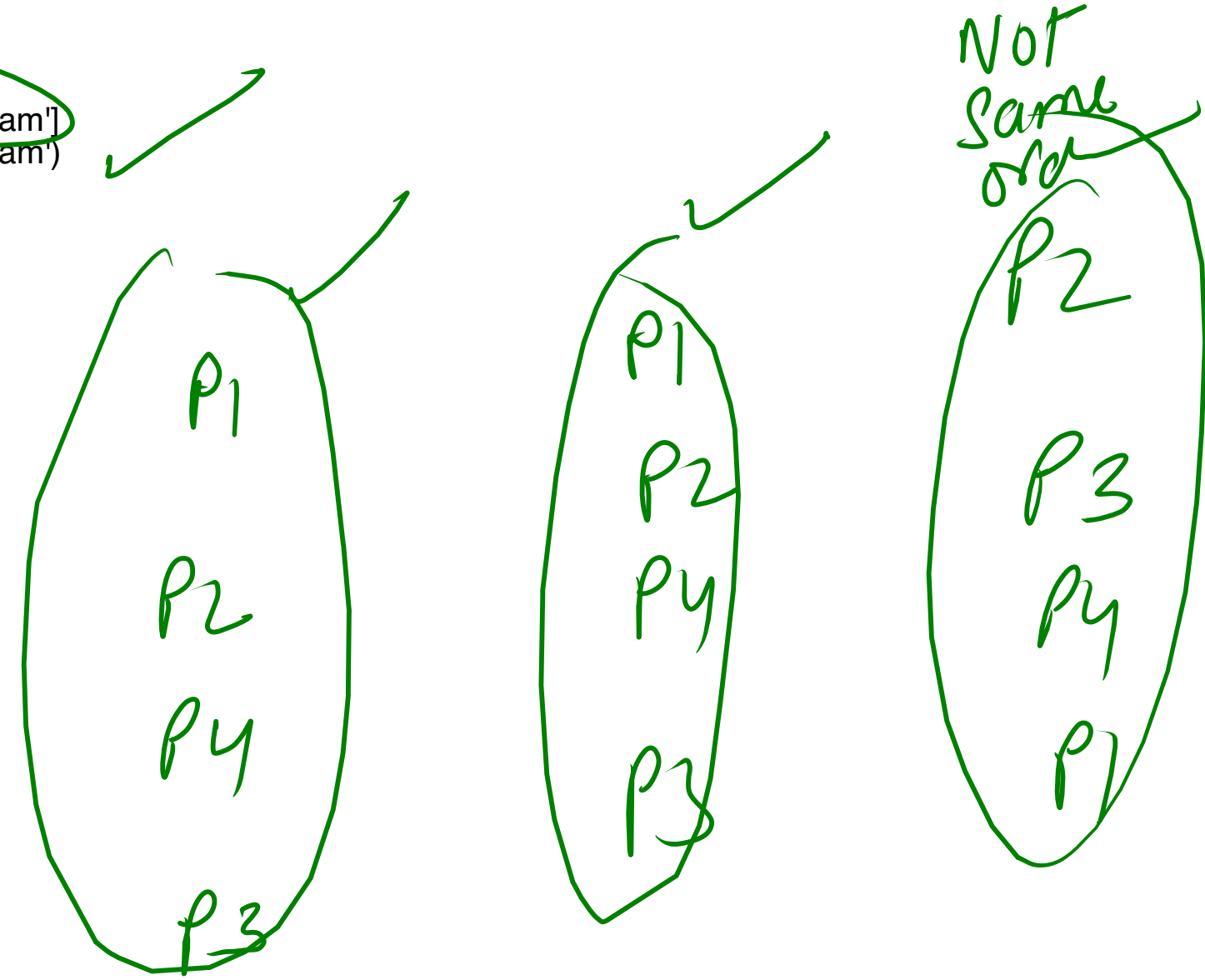
Elements in A but not in B

$B - A = \{4, 5\}$

Elements in B but not in A


Class
A
B
C
B
A
Class (Sports Room)
A
B
C
B
A
Same order
not order
Class (Music)
B
A
C

names in list is ['Akshat', 'John', 'Akshat', 'aman', 'Johnson', 'Adam']
names in tuple ('Akshat', 'John', 'Akshat', 'aman', 'Johnson', 'Adam')
names in set is {'aman', 'Adam', 'Akshat', 'Johnson', 'John'}



Dictionary

Collection of data stored in key-values pairs.

Key : value

✓ name : "Akshat"

age : 35

course : "abc"

decision in code. ← Conditional Statement

If i learn AI then → I will work for google.
else i will work in other technology

If i exercise then i will be fit
else i will be fat

if statement